AYAM BANJADE

CS STUDENT

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BrainTeazer

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ADDITIONAL SKILLS

Rust | C | React | Java | TypeScript | Lua | Kotlin | Git | Shell Script

Linux | Windows | macOS

Docker | Command-line Interface | CMake | Markdown

Financial Mathematics | Stochastic Methods | Technical Writing

LANGUAGES

English:
Native / Bilingual (C2)

German:
Limited working proficiency (A2)

Hindi:
Business fluent

Nepali:
Native / Bilingual (C2)

EDUCATION

Constructor University Bremen — Sept. 2021-Present

B.Sc. Computer Science, Mathematics minor — Bremen, Germany

- Notable Courses: Software Engineering, Database and Web Services, Algorithms and Data Structures, Computer Networks, Operating Systems, Distributed Algorithms, Machine Learning, Secure and Dependable Systems

EXPERIENCE

Constructor University Bremen - Teaching Assistant — Bremen, Germany

Numerical Methods — Feb. 2023 - Present

- Collaborated with peers and the instructor to create and solve challenging problems, and give tutorials on:
  - root finding, interpolation, approximation
  - numerical differentiation, numerical integration (quadrature), and numerical solutions for differential equations

Programming in C/C++ — Sept. 2022 - Dec. 2022

- Debugged and validated code, and gave tutorials to over 200 students regarding:
  - pointers, pointer arithmetic, object-oriented programming, dynamic memory allocation, data structures, templates, macros, header files, makefiles

Introduction to Computer Science — Sept. 2022 - Dec. 2022

- Coordinated tutorials to reduce students' doubts on:
  - **functional programming** (Haskell), **assembly programming**
  - the foundations of computer science, discrete mathematics, Boolean algebra, Hoare's logic
  - combinational and sequential digital circuits

PROJECTS

Rippit

- Developed frontend and backend of a specialized forum website using **jQuery**, and **MySQL** in **MongoDB** and PHP in collaboration with a team using **Version Control System (git)**
- Implemented using **PHP** to connect the database server and website using SQL queries
- Increased functionality (suggestions, maps) using jQuery and the Leaflet library

Bug World

- Collaborated on various sprints to build a full stack simulation game written in **JavaScript**, **HTML**, and **CSS**
- Made a parser, assembler, simulator, and GUI using **OOP**, and created automated test cases using the Jest testing framework
- Wrote and maintained comprehensive documentation with technical specifications

Scrpr

- Implemented an interactive Python Discord bot that scrapes images and metadata from a static website using `bs4`, and `requests` libraries and RegEx

Stochastic Methods Lab

- Developed stochastic models related to financial mathematics using **Python**'s `matplotlib`, `NumPy`, `pandas`, and `SciPy` libraries and CSV manipulation

TheOdinProject

- Designed and built 3 GUI-based applications: etch-a-sketch, calculator and rock paper scissors game using CSS, HTML, and JavaScript

Spam Email Extracter

- Created a program in **C++** to extract 57 attributes based on UCI's Spambase Data Set to help detect spam emails

Bremen, Germany